HEATMAP NODE · PROBLEM STATEMENT

HeatMap Node problem statement

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-11-01	Amish Chadha	Kit 1.9.1 upgrade: users-and-needs table and stakeholder table added for the user requirements (HMN-URS-001); no change to the problem or its scope
0.6	2026-10-02	Amish Chadha	First partner city and data publisher decided by Amish on 2026-10-02; City of Phoenix Office of Heat Response and Mitigation named as the first candidate to approach
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish: sensor head limit $130
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update: budget scope and mounting height per HMN-DDR-001; operating environment aligned with HMN-CAL-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, prior work, open questions)

0.1	2026-09-25	Amish Chadha	Initial scaffold

HeatMap Node problem statement

Cities decide where to plant trees, build shade and open cooling centers with heat data that is too coarse to see the street a person actually walks down. Heat kills in large numbers, it falls hardest on poorer and hotter neighborhoods, and the measure that matters for the body (heat stress, which includes sun and radiant heat from hot surfaces, not only air temperature) is rarely measured at street level at all.

The problem

Heat is one of the deadliest weather hazards. WHO estimates about 489,000 heat-related deaths each year between 2000 and 2019, 45 % of them in Asia and 36 % in Europe (WHO fact sheet). In the summer of 2022 alone an estimated 61,672 people died of heat in 35 European countries (Ballester et al., *Nature Medicine*, 2023).

Heat is not spread evenly across a city. The US EPA reports that daytime air temperatures in urban areas run about 1 to 7 °F (about 0.5 to 4 °C) above outlying areas, and nighttime temperatures about 2 to 5 °F (about 1 to 3 °C) higher (US EPA). Inside cities the differences follow income and history: land surface temperatures in formerly redlined US neighborhoods are about 2.6 °C higher than in other neighborhoods, and up to 7 °C higher in some cities (Hoffman et al., *Climate*, 2020). WHO notes that informal settlements are often hotter than other urban areas (WHO fact sheet).

Three gaps keep cities from acting on this at block scale:

1. **Official stations are sparse and sited away from streets.** Weather stations are placed in open, grassy sites by design, so they report the background climate, not the hot bus stop or market lane.
2. **Satellites see surfaces, not people.** Land surface temperature maps show hot roofs and roads, but the heat a pedestrian feels depends on air temperature, humidity, wind and radiant heat together.
3. **Air temperature alone understates heat stress in the sun.** The standard occupational index, wet bulb globe temperature (WBGT, ISO 7243:2017), weights a black globe temperature and a natural wet-bulb temperature because sun and humidity drive heat strain. Low-cost street sensors seldom include a globe.

Users and context

User	What they need from the data
City heat, climate or resilience officer	Block-level maps of heat stress over a season to target trees, shade (see CoolShade), cool surfaces and cooling centers, and to check whether they worked
Public health department	Near-real-time heat stress by neighborhood during heat waves, to direct outreach to people at risk
Community groups and schools	A node they can build, host and read, and open data they can take to city hall
Outdoor employers (construction, delivery, markets)	Local WBGT estimates to set work and rest cycles
Researchers	A documented, calibrated, repeatable street-level method that others can replicate

Table 1. Users and needs.

User type	Goals and needs	Typical use scenario	Pain points and challenges
City heat, climate or resilience officer	Block-level heat stress maps over a season to site trees, shade and cooling centers, and to check they worked	Reads a season map each spring and a live view on heat-wave days	Stations are sparse and sited away from streets; satellites see surfaces; budgets allow few instruments
Public health department	Near-real-time heat stress by neighborhood in a heat wave, to direct outreach	Watches a dashboard through a heat alert; calls community groups	Air temperature alone understates risk in the sun; data arrive too late or too coarse
Community group or school	A node they can host and read, and open data to take to city hall	Hosts a node on a nearby pole, checks its page weekly	Little money and little technical help; do not trust closed data
Outdoor employer	Local WBGT to set work and rest cycles	Looks at the nearest node before a shift	Has no street-level index; hand instruments are costly and sporadic
Researcher	A documented, calibrated, repeatable street-level method	Collocates a node with a reference station, then compares blocks	Error of low-cost sensors is rarely stated; no record of calibration
Pole owner or utility	Little added load, no drilling, clear liability	Reviews a request to attach, then inspects after wind	Fear of falling parts, corrosion and unauthorised work on the pole

Stakeholders

Table 2. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Amish Chadha (Design Molecule Labs)	Owner and decision maker; publishes the data until a city takes it over	Decisions in HMN-DEC-001
First partner city (candidate: City of Phoenix Office of Heat Response and Mitigation)	Hosts the network; owns or gets permission for poles; uses the maps	Approach after Amish agrees; nothing is agreed
Pole owners (city street lighting, utilities)	Allow clamped attachments at the height proposed	Ask for an attachment standard and a trial pole
Residents and passers-by	Live and walk beneath the nodes; privacy	Explain what is measured; no personal data (PR-012)
FieldNode, CalRig, TwinKit and CityTwin repos of this lab	Shared core, calibration, gateway and open data export	Cross-repo items in docs/REVIEW.md
Standards and regulators (FCC, ISO/WMO guidance)	Radio, product and measurement practice	Register in HMN-REG-001
Funders and NGOs	Pay for networks in low-income districts	Cost per node in HMN-MKT-001

Operating environment

- **Mounting:** existing street light, sign or utility poles (60 to 200 mm in diameter), with the asset owner's permission, with the sensor arm at about 2.8 m above the pavement (HMN-DDR-001 D2), pointing toward the

equator where the street allows (HMN-DDR-002). HMN-CAL-001 estimates that on a sunny afternoon the air at pedestrian height is 0.2 to 0.8 °C warmer than at the sensors.

- **Climate:** air temperature from about -20 to +50 °C at the node (R8, restated to match under HMN-DDR-002), full sun, rain, dust and salt air near coasts; black globe temperatures can exceed 60 °C in full sun (estimate).
- **Connectivity:** LoRaWAN coverage from a community gateway (for example the lab's TwinKit gateway) or a public network; no mains power at the node.
- **Density:** tens of nodes per neighborhood, so each must be cheap, quick to fit and self-powered.

Constraints

- Garage-buildable prototype, $130 USD or less for the HeatMap-specific sensor head (raised from $120 by a budget top-up Amish approved on 2026-09-26). The FieldNode core, its sun shield and the pole adapter that seats it on street poles are counted against FieldNode, a shared lab component costed and budgeted in its own repo (HMN-DDR-001 D1 and HMN-DDR-002, decided by Amish on 2026-09-25). HMN-CAL-001 v0.3 prices the head at $127 with the aspiration fan, within this limit.
- Built on the FieldNode core (enclosure, solar, battery, LoRaWAN radio) so that no power or radio design is repeated.
- Privacy: environmental measurements only. No camera, microphone or device tracking.
- No drilling of poles; fit with stainless band clamps.
- Open hardware (CERN-OHL-S-2.0) and open software (MIT); data in an open format.

Out of scope

- Personal heat stress for an individual (wearables) and indoor heat.
- Issuing official heat warnings. The node supplies data; public warnings stay with the responsible agency.
- Air quality, noise and traffic, which are covered by the AirStreet, NoiseMap and CurbCount concepts.

Prior work

- **WBGT standard and models.** ISO 7243:2017 defines WBGT as a screening index for heat stress (ISO). Liljegren and colleagues showed how to estimate WBGT from standard weather measurements, including globe temperature and wind speed (Liljegren et al., *J. Occup. Environ. Hyg.*, 2008); an open implementation of their code is available (wbgt package). HeatMap Node measures the inputs this model needs.
- **Heat and equity studies.** Satellite studies across 175 large US urbanized areas found that the average person of color lives in a census tract with higher surface urban heat island intensity than non-Hispanic white residents in all but 6 of them (Hsu et al., *Nature Communications*, 2021). These studies use surface temperature; HeatMap Node adds what people experience at street level.
- **Mobile heat campaigns.** Some cities map heat on a single hot day with vehicle-mounted or hand-held sensors. These give good spatial detail but not the season-long, day-and-night record that fixed nodes give.
- **Lab components.** FieldNode (shared outdoor core), CalRig (sensor calibration chamber) and TwinKit (gateway and data layer) are sibling concepts in this lab. CoolShade is the lab's shade structure concept that HeatMap data is meant to help site.

Open questions

- Which city or community partner hosts the first network, and on whose poles? Decided 2026-10-02: a hot US city with a dedicated heat office; the first candidate to approach is the City of Phoenix Office of Heat Response and Mitigation, using the US915 band. Nothing is agreed, and whose poles is still to be settled with the city (HMN-DEC-001).
- What mounting height is acceptable to pole owners? HMN-CAL-001 gives a first estimate of the height error (up to about 0.8 °C in strong sun); a field comparison of 2.0 and 2.8 m is still needed.
- Which output do users want most: air temperature, WBGT, mean radiant temperature, or a simple heat stress category?
- Who owns and publishes the data? Decided 2026-10-02: Amish's lab publishes it under an open licence from CityTwin's open data export until the partner city takes it over (HMN-DEC-001).

User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [ ] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [ ] Run co-design sessions with intended users; record who, where and what was learned
- [ ] Validate load, distance, terrain and cost assumptions in the field
- [ ] Revise requirements (REQ) from findings before freezing the design